

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Deep Learning

COURSE CODE: MLA0408

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 5

Duration: 1 Hour
Total Marks:50

Annexure A

Analytical Problem Solving

1. A coin is tossed 10 times and the outcomes are:
H H T H H H T H H T
Using Maximum Likelihood Estimation, estimate the probability of getting Heads (p).
2. Out of 50 students, 42 passed an exam.
Assuming a Bernoulli distribution, find the MLE of the probability that a student passes.
3. A factory produces light bulbs. In a sample of 100 bulbs, 8 are defective.
Find the MLE for the probability that a bulb is defective.
4. A biased die is rolled 60 times. The number 6 appears 18 times.
Estimate the probability of rolling a 6 using MLE.
5. Given:
Input (x) = 4
Weight (w) = 2
Actual Output = 10
Prediction:
 $y = w \times x$
 1. Find the predicted output.
 2. Calculate the error.
 3. If the gradient is 4 and learning rate is 0.01, find the updated weight.

Code of Conduct:

*I **G. Naveen Kumar Reddy** (192525288) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

G. Naveen Kumar Reddy

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

1. A coin is tossed 10 times and the outcomes are:

H H T H H H T H H T?

Given the outcomes:

H H T H H H T H H T

Step 1: Count the outcomes

- Total tosses $n=10$
- Number of Heads $H=7$
- Number of Tails $T=3$

Step 2: MLE formula

For a coin, the Maximum Likelihood Estimate of the probability of Heads is:

$\hat{p} = \frac{\text{Number of Head}}{\text{Total Tosses}}$

$\hat{p} = \frac{7}{10} = 0.7$

2. Out of 50 students, 42 passed an exam. Assuming a Bernoulli distribution, find the MLE of the probability that a student passes.

Given:

- Total students = 50
- Students who passed = 42
- Students who failed = $50 - 42 = 8$

MLE Formula

For a Bernoulli distribution, the MLE of probability of success is:

$\hat{P} = \frac{\text{Number of successes}}{\text{Total Observations}}$

Here, **passing = success**.

$$\hat{P}=42/50$$

$$\hat{P}=0.84$$

3. A factory produces light bulbs. In a sample of 100 bulbs, 8 are defective. Find the MLE for the probability that a bulb is defective.

Given:

- Total bulbs = 100
- Defective bulbs = 8
- Non-defective bulbs = 92

MLE Formula

For a Bernoulli distribution:

$$\hat{p}=\text{Number of defective bulbs}\backslash\text{Total bulbs}$$

Substitute the values:

$$\hat{p}=8/100$$

$$\hat{p}=0.08$$

4. A biased die is rolled 60 times. The number 6 appears 18 times. Estimate the probability of rolling a 6 using MLE.

Given:

- Total die rolls = 60
- Number of times 6 appears = 18

MLE Formula

$$\hat{p}=\text{Number of successes}\backslash\text{Total trials}$$

Here, rolling a 6 is considered a success.

$$\hat{p}=18\backslash60$$

$$\hat{p}=0.3$$

5. Given:

Input (x) = 4

Weight (w) = 2

Actual Output = 10

Prediction:

$$y = w \times x$$

1. Find the predicted output.

2. Calculate the error.

3. If the gradient is 4 and learning rate is 0.01, find the updated weight.

Given:

- Input $x=4$
- Weight $w=2$
- Actual output $=10$
- Gradient $=4$
- Learning rate $\eta=0.01$

1. Predicted Output

Formula:

$$y = w \times x$$

$$y = 2 \times 4 = 8$$

So, Predicted Output = 8.

2. Calculate the Error

Using:

$$\text{Error} = \text{Actual} - \text{Predicted}$$

$$\text{Error} = 10 - 8 = 2$$

So, Error = 2.

3. Updated Weight

Gradient descent formula:

$$W_{\text{new}} = w_{\text{old}} - (\text{Learning Rate} \times \text{Gradient})$$

$$W_{\text{new}} = 2 - (0.01 \times 4)$$

$$= 2 - 0.04$$

$$W_{\text{new}} = 1.96$$